LOST & FOUND MANAGEMENT

A PROJECT REPORT

Submitted by

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(241401013)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

Computer Science & Business System



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "LOST & FOUND MANAGEMENT" is the bonafide work of **AKASH SUNDAR (241401013)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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I, **AKASH SUNDAR (241401013)**, hereby declare that the project report titled "LOST & FOUND MANAGEMENT", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman **Mr. S. Meganathan, B.E., F.I.E.,** Chairperson **Dr. Thangam Meganathan, Ph.D.,** and Vice-Chairman **Mr. Abhay Shankar Meganathan, B.E., M.S.,** for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, **Dr. S. N. Murugesan, M.E., Ph.D.,** Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank **Dr. K. R. Sowmia, Ph.D.,** Associate Professor & Head, Department of Computer Science and Business Systems, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, **M. L. Alphin Ezhil Manuel,** Assistant Professor (SG), for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

AKASH SUNDAR

ABSTRACT

Lost property is a routine but poorly managed problem on educational campuses. Students and staff misplace identity cards, keys, books, electronics and accessories every day, and the usual response is an informal chain of enquiries: messages in group chats, notices on boards and visits to the security desk or administration office, with details noted in a paper register or a spreadsheet. Such a process has no single searchable record, no way to see whether an item has already been claimed, no verification of ownership before an item is released and no automatic communication with the people involved.

This project presents Lost & Found Management, a business application developed on the Zoho Creator low-code platform to digitise the complete life cycle of a lost or found item. The application contains three core forms, Item Reports, Claims and Handover Records, linked through lookup fields, together with eleven list reports, a Campus Dashboard page with quick-action buttons and two charts, and eleven form workflows. Ten Deluge scripts enforce the business rules: a claim cannot be raised against an item that is already resolved, ownership evidence is compulsory, a handover can be recorded only against an approved claim and contact details are checked before submission. A Send Email action informs the administrator and the reporter of every new report, and a Deluge sendmail task alerts the assigned verifier when a claim is submitted. Access is controlled through Zoho Creator profiles and roles, and personal fields such as the claimant name and contact e-mail are flagged as personal data.

Testing was organised at unit, integration and role-based levels, with test cases mapped to the functional requirements. Performance is kept predictable through criteria-based reports, thumbnails for item photographs and single-record fetches in scripts. By replacing registers and ad-hoc messages with a verified claim-and-handover workflow, the application gives the campus a searchable, auditable and notification-driven lost-and-found service that is available on the web and in the Zoho Creator mobile app, and it was built without a conventional codebase.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Lost and Found Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	LOST & FOUND MANAGEMENT
Application Domain	Education / campus administration (lost-and-found property management)
Student Name & Register No.	AKASH SUNDAR – 241401013
Zoho Creator Edition	Zoho Creator trial edition
Application Name (Link Name)	LOST&FOUND (link name: lost-found)
Published / Live URL	https://athinarayananakashsundar1278.zohocreatorportal.in
Demo Credentials (per role)	[Administrator: ____ | Verifier (Admin Access): ____ | Reporter: ____]
Number of Forms / Subforms	4 / 0 (3 core forms + 1 starter-template sample form, Add Order)
Number of Reports (by type)	List: 11 | Kanban: 0 | Calendar: 0 | Pivot/Chart: 0 (2 charts on the dashboard page) | Others: 0
Number of Pages / Dashboards	1 (Campus Dashboard)
Number of Workflows	Form: 11 | Schedule: 0 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	10 custom Deluge scripts + 1 Send Email action; 0 user-defined functions
Integrations	E-mail notifications through the built-in Zoho Creator mail service (sendmail / Send Email); no external API
Roles & Profiles Defined	Profiles: Administrator, Admin Access (custom), Read, Write, Customer (portal); roles: CEO > Admin
Environments Used	[Development → Stage → Production – state the environments used]
Date of Production Deployment	[DD Month YYYY]

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DS	Deluge Script (script identifier)
ER	Entity–Relationship
FR	Functional Requirement
HTML	HyperText Markup Language
LCDP	Low-Code Development Platform
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

A college campus is a small town. Thousands of students, faculty members and staff move through classrooms, laboratories, libraries, hostels and canteens every day. Along the way, items are misplaced – identity cards, keys, calculators, textbooks, earphones, water bottles and clothing – and other members of the community find them and want to return them. The security desk and the administrative office normally receive such items, but matching a found item with its owner depends on memory, personal contacts and handwritten records.

The business process addressed in this project is the complete lost-and-found cycle of a campus: (i) a person reports a lost item or a found item with its description, category, location and date; (ii) anyone who believes an item is theirs raises a claim with evidence of ownership; (iii) a verifier reviews the claim and approves or rejects it; and (iv) an approved claim ends in a recorded handover, after which the report can be closed as resolved. The people involved are the reporters and claimants (students and staff), the verifiers who work at the security or administration desk, and an administrator who oversees the service.

1.2 Problem Statement

Campuses lack a single, structured place to record lost and found items and to manage the claims made against them. As a result, owners cannot easily discover that their item has been found, finders do not know where to hand an item over, and staff cannot confirm that an item is being returned to its rightful owner. The absence of a verified claim-and-handover trail also exposes the institution to disputes and misuse.

- Information is scattered across registers, spreadsheets, notice boards and messaging groups, so there is no single searchable record of lost and found items.
- Ownership is rarely verified formally; items may be released on a verbal description alone and the decision is not recorded.
- Status is invisible: nobody can tell whether a reported item is still open, under review or already resolved, which leads to duplicate or invalid claims.

- Staff and reporters are not notified automatically, and there is no dashboard to summarise reports by category or type.

1.3 Objectives

- To design a normalised data model and forms for item reports, ownership claims and handover records, linked through lookup fields.
- To automate claim validation, default status assignment and e-mail notification using Deluge scripting and form workflows.
- To provide role-based dashboards and reports that let administrators, verifiers and reporters browse lost, found, pending and resolved items.
- To notify the administrator, reporter and assigned verifier by e-mail through the built-in Zoho Creator mail service at the key events of the process.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

In scope: a web application, also usable through the Zoho Creator mobile app, for a single campus. It covers (a) reporting of lost and found items with an optional photograph; (b) browsing of item reports through filtered reports for all, lost, found and resolved items; (c) submission of ownership claims against an item report; (d) verification of claims by an assigned verifier through the Pending Claims and Claims Under Review reports; (e) recording of the final handover; (f) a Campus Dashboard that summarises reports; and (g) role-based access through profiles and roles.

Out of scope: automatic image-based matching of lost and found items, integration with the campus ID-card or student-information system, SMS or WhatsApp notifications, reward or payment handling and multi-campus deployment. These are listed as future enhancements in Section 9.2. The sample form Add Order, which came with the starter structure of the application, is retained but is not part of the lost-and-found process.

1.5 Justification for a Low-Code Approach

Low-code platforms let applications be assembled largely through visual configuration of forms, reports and workflows, with code written only where the business logic demands it [1]. For a campus lost-and-found service this approach is well suited. The data model is small and form-centred, and the users are non-technical. Zoho Creator

supplies hosting, security, a mobile app and report and dashboard builders out of the box [2], so the project could concentrate on the business process rather than on infrastructure. The remaining logic, namely validations, default values and notifications, maps directly to Deluge form events [3].

Compared with a custom web application built with a front-end framework, an application server and a database, development time, hosting effort and maintenance cost are lower, and the application inherits the access control and audit capabilities of the platform. Changes such as a new item category or an extra claim status can be made by a citizen developer in the form builder without a separate deployment cycle. The trade-offs, namely limited user-interface customisation and dependence on the vendor platform, are acceptable for an institutional service of this size and are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing arrangement, lost-and-found handling on a campus is informal and manual. A person who loses an item asks classmates, posts a message in a messaging group or on a notice board, and may visit the security desk or administration office. A person who finds an item usually hands it to the security desk, where the details are written in a paper register or a spreadsheet. Matching is done from staff memory or by scanning the register when someone enquires, and the item is released after a verbal description by the owner, followed by a signature in the register. Figure 2.1 summarises this as-is process.

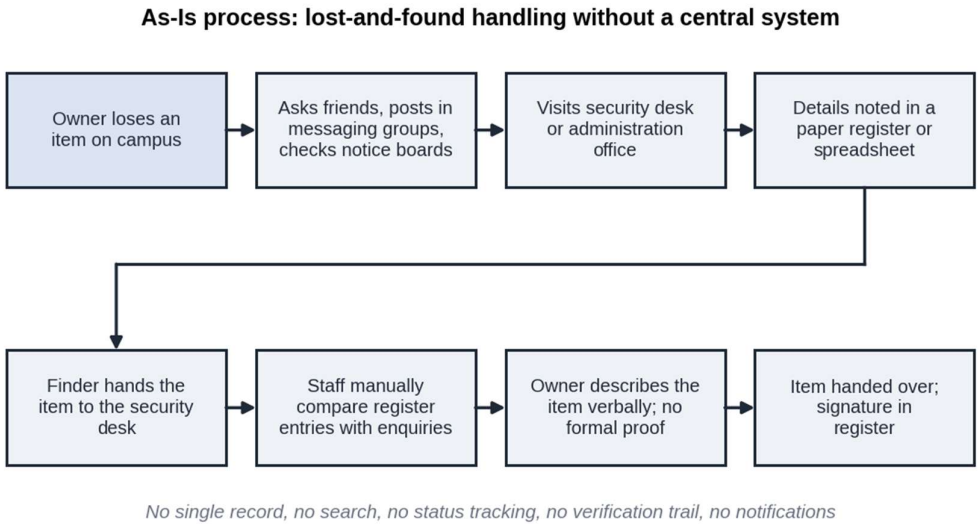


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- No central, searchable record: entries are spread across registers, spreadsheets and messages.
- No ownership verification trail: release depends on a verbal description and is not recorded against a reviewed claim.
- No visibility of status: owners cannot see whether an item has been found, is under review or has been handed over.

- No automatic notification: staff and reporters learn about new reports and claims only by chance or by enquiry.
- No reporting or analysis: management cannot see which categories of items are lost most often or how many cases remain open.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 6–8 weeks for a small team: requirement analysis, database design, back-end and front-end coding, testing and deployment	Estimated 1–2 weeks for one developer: forms, reports, scripts and dashboard configured in the builder
Skills required	Full-stack programming (HTML/CSS/JavaScript, a back-end language, SQL) and server administration	Form and report configuration plus basic Deluge scripting
Hosting & infrastructure	Web server, database server, domain, SSL and backups must be arranged and maintained by the college	Hosted, backed up and secured on the Zoho cloud; published from the builder
Mobile application	Separate mobile app or a responsive redesign is needed	Available in the Zoho Creator mobile app, including camera access for the item photo field
Security & access control	Authentication, authorisation, input validation and data protection coded and tested by the developer	Profiles, roles, mandatory fields, personal-data flags and Deluge validations provided by the platform
Maintenance & change	Every change needs coding, testing and redeployment	New fields, statuses and reports are added in the builder without redeployment
Customisation limits	Practically unlimited	Bounded by platform components, report layouts and Deluge execution limits
Cost	Developer time, server hosting and licences for the technology stack	Platform subscription or education access; no server cost and far less developer time

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide	Introduces low-code principles, visual development and how business applications are built with minimal hand-coding	Basis of the low-code justification in Section 1.5
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, roles and deployment in Zoho Creator	Primary reference for building and configuring the application
3	Zoho Corporation [3]	Deluge Script Reference Guide	Defines Deluge syntax: fetch, conditions, loops, sendmail and form events	Used for scripts DS-01 to DS-10
4	Zoho Corporation [4]	Zoho Creator Learning Centre	Guided learning material and sample applications for Zoho Creator	Used to learn report, dashboard and workflow configuration
5	A. Sahay et al., 2020 [5]	Supporting the understanding and comparison of low-code development platforms	Proposes a conceptual framework and compares eight low-code platforms by their features	Gives the feature-based view of low-code platforms used to justify the platform choice
6	D. Di Ruscio et al., 2022 [6]	Low-code development and model-driven engineering: Two sides of the same coin?	Examines how low-code relates to model-driven engineering	Explains the model-driven idea behind configuring forms and workflows instead of coding them
7	S. Käss et al., 2023 [7]	Practitioners' perceptions on the adoption of low code development platforms	Reports practitioners' views on the drivers and barriers of adopting low-code platforms	Supports the benefits and limitations discussed in Sections 1.5 and 8.4

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
8	Y. Luo et al., 2021 [8]	Characteristics and challenges of low-code development: The practitioners' perspective	Analyses Stack Overflow and Reddit posts; low-code is favoured for automated processes and workflows, with mixed views on its pros and cons	Relevant because this application is a workflow-centred business process
9	A. Sahay et al., 2023 [9]	Analyzing business process management capabilities of low-code development platforms	Compares how eight low-code platforms specify business processes and workflows	Supports the workflow-based design in Section 4.4

2.5 Emerging Trends in Low-Code Development

Practitioner studies report that low-code is favoured where processes and workflows must be automated [8] and that adoption is driven by development speed and the involvement of business users [7]. Four trends are relevant to this project. First, citizen development: a campus administrator can maintain the application without a software team, which influenced the choice of simple forms and descriptive workflow names. Second, hyper-automation: validations, default values and e-mail notifications remove manual follow-up from the lost-and-found process. Third, composable applications: the three forms and their reports can be extended with new modules, such as a portal or an analytics view, without redesigning the data model. Fourth, AI-assisted development and AI fields such as image recognition are becoming standard platform features; they were not required for this version and are listed as a future enhancement.

2.6 Proposed System

The proposed system is a Zoho Creator application named LOST&FOUND that overcomes each limitation listed in Section 2.2. The Item Reports form and its reports provide one central, searchable record (limitation 1). The Claims form, with compulsory ownership evidence and a Pending → Under Review → Approved/Rejected status, and the rule that a handover is accepted only for an approved claim, create a verification trail (limitation 2). Status fields on item reports, claims and handovers, with filtered reports, make progress visible (limitation 3). E-mail

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Campus lost-and-found administrator who owns the application	Manage users, profiles and settings; view all data; receive an e-mail for each new item report; monitor the dashboard
Verifier	Security or administration desk staff using the Admin Access profile	Review claims; set claim status to Under Review, Approved or Rejected; record review notes; record handovers; update item report status
Reporter / Claimant	Student or staff member who has lost or found an item (Customer / portal profile)	Report lost or found items with a photograph; browse reports; submit ownership claims with evidence

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a user to report a lost or found item by entering item name, report type, description, category, date, location, reporter name and contact e-mail, with an optional photograph.	Reporter	High
FR-02	The system shall check the structure of the contact e-mail address when an item report is submitted.	Reporter	Medium
FR-03	The system shall set the status of a new item report to Open and allow it to move to Under Review and Resolved.	Verifier, Administrator	Medium
FR-04	The system shall send an e-mail notification to the administrator and the reporter when a new item report is added.	System	Medium
FR-05	The system shall provide filtered reports of all, lost, found and resolved items sorted by date, with resolved items excluded from the lost and found lists.	All roles	High
FR-06	The system shall allow a claimant to submit an ownership claim against an item report with claimant name and ownership evidence.	Claimant	High
FR-07	The system shall reject a claim if no item report is selected, if ownership evidence is empty or if the item report is already Resolved.	Claimant, System	High

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall set the status of a new claim to Pending.	System	Medium
FR-09	The system shall send an e-mail to the assigned verifier when a new claim is submitted.	System	Medium
FR-10	The system shall allow a verifier to update a claim to Under Review, Approved or Rejected and to record review notes.	Verifier	High
FR-11	The system shall provide Pending Claims and Claims Under Review reports to verifiers.	Verifier	Medium
FR-12	The system shall allow a handover to be recorded only against a selected item report and an Approved claim.	Verifier	High
FR-13	The system shall record the verifier, handover date, status and resolution notes of a handover and list completed handovers.	Verifier	High
FR-14	The system shall display a Campus Dashboard with quick-action buttons and charts of reports by category and by report type.	Administrator, Verifier	Medium
FR-15	The system shall restrict access to forms, reports and pages according to the user's profile.	Administrator	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Filtered reports of up to 1,000 item reports shall open within 3 seconds on a normal broadband connection (target)
NFR-02	Security	Users shall access only the forms, reports and pages permitted by their profile
NFR-03	Usability	The application shall be usable on the web and in the Zoho Creator mobile app, with camera or gallery upload for item photos
NFR-04	Availability	The application shall be available through the Zoho cloud without the college maintaining servers
NFR-05	Maintainability	Components shall use consistent link names and descriptive workflow names so that a citizen developer can modify them
NFR-06	Auditability	Record creation time and user shall be traceable through system fields and the platform audit trail
NFR-07	Privacy	Personal fields (claimant name, contact e-mail) shall be flagged as personal data

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator trial edition (creator.zoho.in data centre)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used (no REST integration in this version)
Other Zoho services	Zoho Creator built-in mail service (sendmail / Send Email); no other Zoho service
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

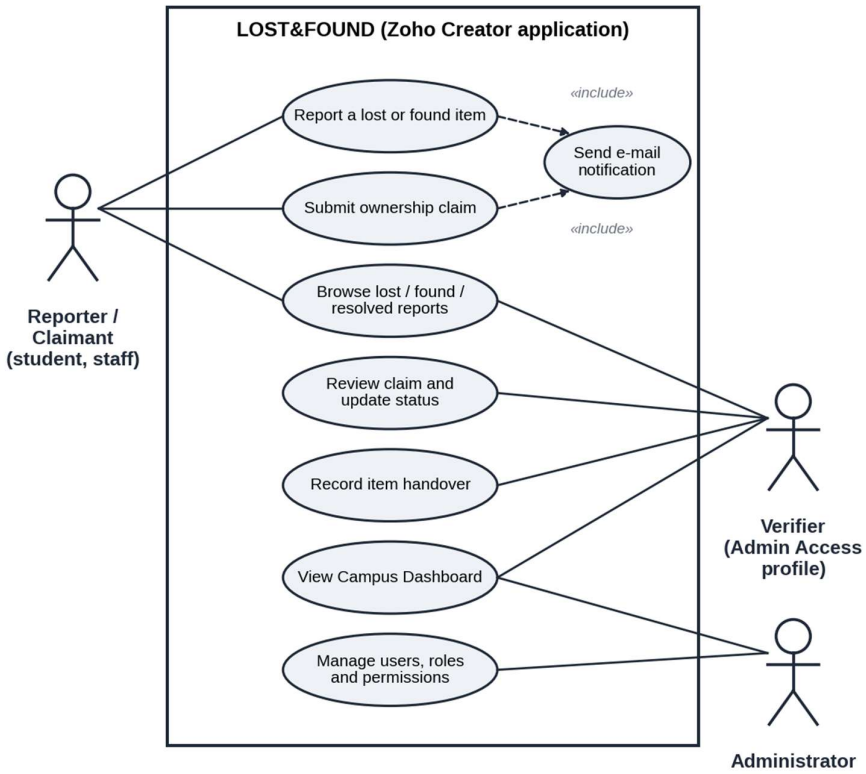


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Report a Lost or Found Item

Use Case ID	UC-01
Use Case Name	Report a Lost or Found Item
Actor(s)	Reporter / Claimant
Pre-condition	The user can open the application and the Item Reports form
Main Flow	1. Open the Campus Dashboard and click “Report a Lost or Found Item”. 2. Select the report type (Lost or Found). 3. Enter item name, description, category, date, location, reporter name and contact e-mail; optionally add a photograph. 4. Click Submit. 5. The system checks mandatory fields and the e-mail format (DS-10). 6. The record is saved with status Open and a success message is shown. 7. An e-mail is sent to the administrator and the reporter (DS-11).
Alternate Flow	5a. A mandatory field is empty or the e-mail format is wrong: a message is shown and the user corrects the entry.
Post-condition	The item report is stored with status Open and appears in the Lost Items or Found Items report
Related FR	FR-01 to FR-05

Table 3.6: Use Case Description – Submit and Verify an Ownership Claim

Use Case ID	UC-02
Use Case Name	Submit and Verify an Ownership Claim
Actor(s)	Claimant, Verifier
Pre-condition	An item report exists and its status is not Resolved
Main Flow	1. The claimant opens the Claims form; Claim Status defaults to Pending (DS-04). 2. The claimant selects the Item Report, enters name and ownership evidence, and the assigned verifier's e-mail. 3. On Submit the system runs DS-01, DS-02 and DS-05. 4. The claim is saved and the verifier is e-mailed (DS-03). 5. The verifier opens Pending Claims, checks the evidence and sets the status to Under Review. 6. The verifier records Review Notes and sets the status to Approved or Rejected.
Alternate Flow	3a. The item report is Resolved, no item report is selected or evidence is empty: an alert is shown and submission is cancelled. 4a. No verifier is assigned: no e-mail is sent and an information message is logged.
Post-condition	The claim is stored with its updated status and is listed in the matching claim report
Related FR	FR-06 to FR-11

Table 3.7: Use Case Description – Record Item Handover

Use Case ID	UC-03
Use Case Name	Record Item Handover
Actor(s)	Verifier
Pre-condition	A claim with status Approved exists for the item report
Main Flow	1. Open the Handover Records form (dashboard button "Record Item Handover"). 2. Handover status is set to Completed on load (DS-07). 3. Select the Item Report and the Approved Claim; enter Verified By, Handover Date and Resolution Notes. 4. Click Submit; DS-06, DS-08 and DS-09 validate the references. 5. The handover record is saved and appears in Completed Handovers. 6. The verifier sets the Item Report status to Resolved.
Alternate Flow	4a. The claim is missing, not found or not Approved: an alert is shown and the handover is not saved.
Post-condition	The handover is recorded and the item report can be closed as Resolved
Related FR	FR-12, FR-13

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

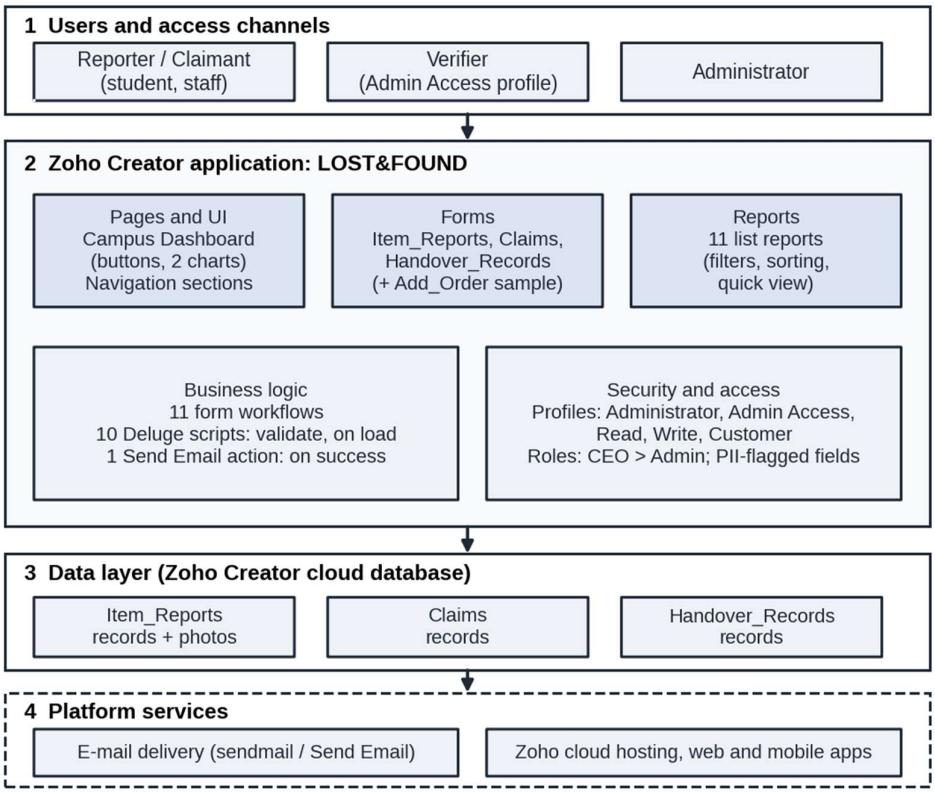


Figure 4.1: System Architecture of the Application

Figure 4.1 shows the layered architecture of the application. The first layer contains the users and their access channels: reporters and claimants, verifiers and the administrator reach the application through a web browser or the Zoho Creator mobile app after signing in.

The second layer is the LOST&FOUND application itself. The presentation part consists of the Campus Dashboard page, the navigation sections and eleven list reports. The data-capture part consists of the Item Reports, Claims and Handover Records forms (and the starter Add Order form). The business-logic part holds eleven form workflows: ten Deluge scripts that validate input or set default values and one Send Email action. Security is provided by profiles, roles and personal-data flags.

The third layer is the Zoho Creator cloud database in which every form stores its records, including the uploaded item photographs. The fourth layer comprises platform services: the mail service used by sendmail and the Send Email action, and the Zoho cloud hosting that serves the web and mobile applications. No external API is called in this version.

4.2 Data Model

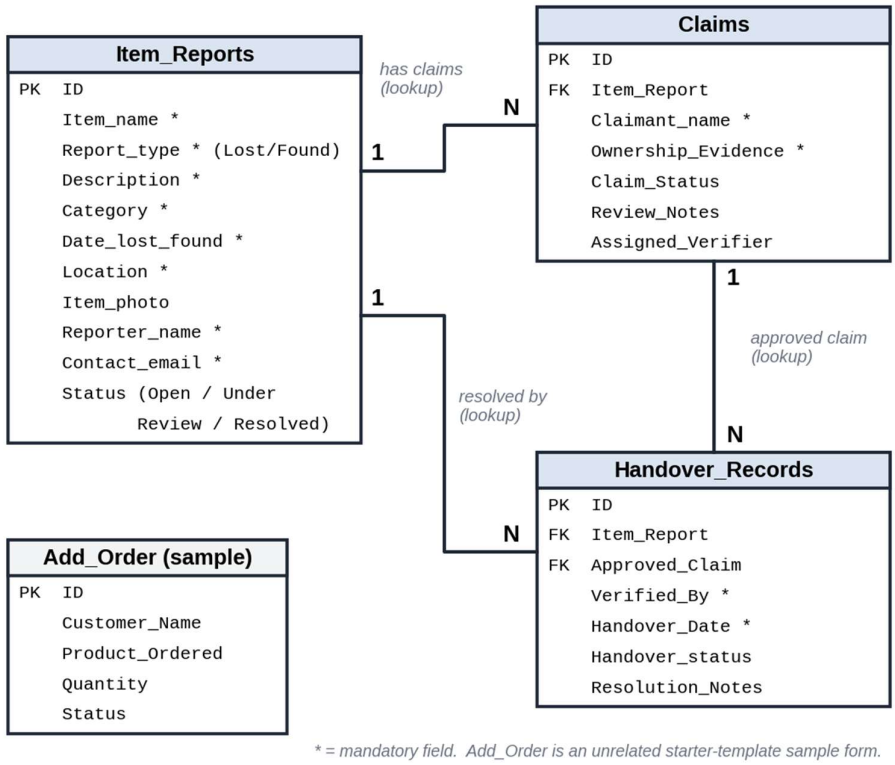


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Item_Reports	Item_name, Report_type, Category, Date_lost_found, Status	Claims	One-to-Many	Lookup field Item_Report in Claims

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Item_Reports	Item_name, Report_type, Category, Date_lost_found, Status	Handover_Records	One-to-Many	Lookup field Item_Report in Handover_Records
Claims	Claimant_name, Ownership_Evidence, Claim_Status	Handover_Records	One-to-Many	Lookup field Approved_Claim in Handover_Records
Add_Order (sample)	Customer_Name, Product_Ordered, Order_Date, Status	None	Stand-alone	Starter-template form

Item_Reports is the central entity: every claim and every handover refers to exactly one item report. Claims.Item_Report is a lookup to Item_Reports, so one item report can receive many claims. Handover_Records holds two lookups: Item_Report points to the reported item, and Approved_Claim points to the claim that justified the handover. Zoho Creator exports lookup fields as picklists that take their values from the ID of the parent form, which is how they appear in the application definition.

No subform is used. Each claim and each handover is a record of its own so that they can be reported, filtered and audited independently. The Add Order form from the starter structure is unrelated to the other forms.

4.3 Form Design

Table 4.2: Field Specification – Item_Reports Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Item name	Item_name	Single Line	Yes	—	Name of the lost or found item
Report type	Report_type	Radio	Yes	Lost / Found	Separates lost from found items
Description	Description	Multi Line	Yes	—	Identifying details such as colour, brand, marks
Category	Category	Drop Down	Yes	Electronics, ID cards, Keys, Books & Notes, Accessories, Clothing, Other	Used by the dashboard column chart
Date lost/found	Date_lost_found	Date	Yes	—	Sort key of the item reports
Location	Location	Single Line	Yes	—	Place of loss or finding

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Item photo	Item_photo	Image	No	Camera or gallery upload	Shown as a thumbnail in reports
Reporter name	Reporter_name	Single Line	Yes	—	Person who reports the item
Contact email	Contact_email	Email	Yes	Maximum 80 characters; format check DS-10	Flagged as personal data; receives the notification e-mail
Status	Status	Drop Down	No	Open / Under Review / Resolved; default Open	Drives the Resolved and Exclude Resolved filters

Table 4.3: Field Specification – Claims Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Claimant name	Claimant_name	Name	Yes	First and last name	Flagged as personal data
Item Report	Item_Report	Lookup (Item_Reports)	Via DS-02	Shows Item name; sorted ascending	Links the claim to the reported item
Ownership Evidence	Ownership_Evidence	Multi Line	Yes	Checked again by DS-05	Proof such as marks, contents or receipts
Claim Status	Claim_Status	Drop Down	No	Pending / Under Review / Approved / Rejected; default Pending (DS-04)	Updated by the verifier
Review Notes	Review_Notes	Multi Line	No	—	Verifier's remarks
Assigned Verifier	Assigned_Verifier	Single Line	No	E-mail address used by DS-03	If empty, no notification is sent

Table 4.4: Field Specification – Handover_Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Item Report	Item_Report	Lookup (Item_Reports)	Via DS-09	Shows Item name; sorted ascending	Item being handed over

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Approved Claim	Approved_Claim	Lookup (Claims)	Via DS-06, DS-08	Only an Approved claim is accepted	Claim that justifies the handover
Verified By	Verified_By	Single Line	Yes	—	Staff member who handed the item over
Handover Date	Handover_Date	Date	Yes	—	Date of the physical handover
Handover status	Handover_status	Drop Down	No	Completed / Pending; set to Completed on load (DS-07)	Drives the Completed Handovers report
Resolution Notes	Resolution_Notes	Multi Line	No	—	Remarks about the handover

The Add Order form is a starter-template sample and is not specified here because it takes no part in the lost-and-found process. Field-level validation is provided by the mandatory flags above and by the Deluge scripts listed in Table 5.2.

4.4 Process and Workflow Design

The lost-to-handover process has three connected flows, shown in Figure 4.3. In the item-report flow, a reporter submits an Item Report; DS-10 checks the e-mail format, the record is saved with status Open and DS-11 e-mails the administrator and the reporter.

In the claim flow, a claimant submits a Claim; DS-04 defaults the status to Pending, and on submission DS-01, DS-02 and DS-05 reject the claim if the item is already Resolved, no item is selected or evidence is empty. After saving, DS-03 e-mails the assigned verifier, who reviews the claim and sets it to Under Review, Approved or Rejected.

In the handover flow, the verifier opens the Handover Records form; DS-07 sets the status to Completed, and DS-06, DS-08 and DS-09 allow the record to be saved only if an item report and an Approved claim are selected. Scripts run on three events: On Load (defaults), On Validate (rules) and On Success (notifications).

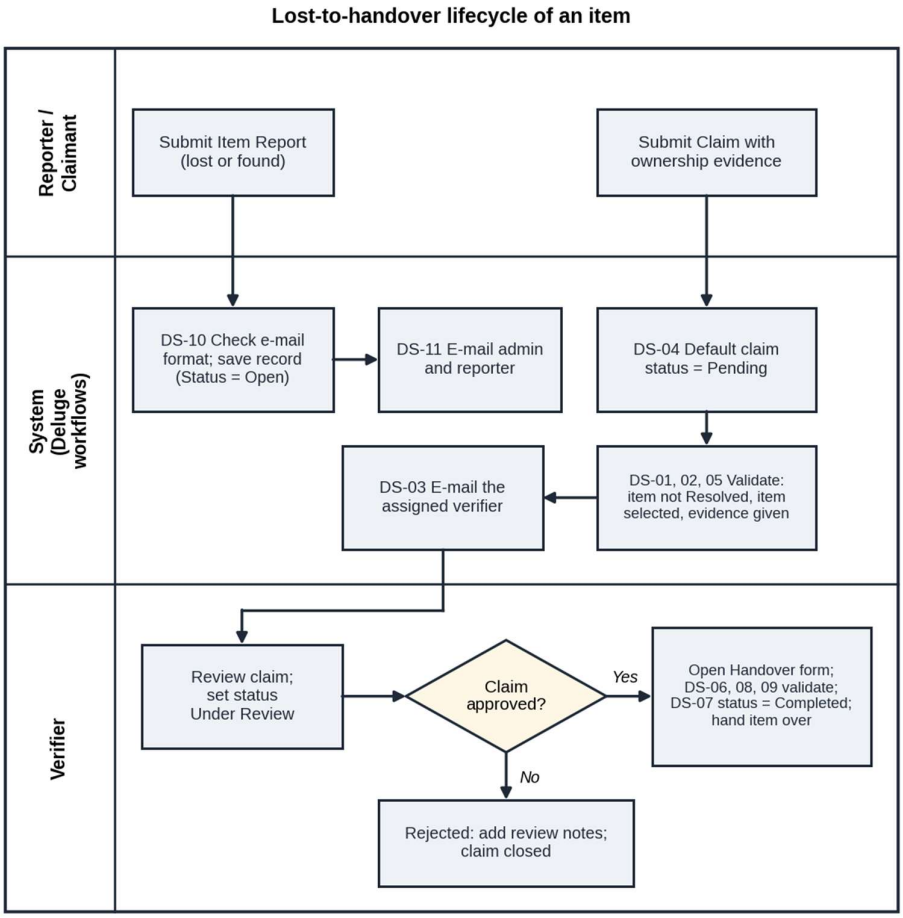


Figure 4.3: Workflow / Process Flow for the Lost-to-Handover Lifecycle

4.5 User Interface and Navigation Design

Figure 4.4 shows the navigation structure and the wireframe of the Campus Dashboard. The menu opens on Home (Campus Dashboard) followed by sections for Item Reports, Claims and Handover Records, each holding its form and its reports. The starter sections All Orders and Add Order, and the system sections Reports, Approvals and App Preferences, are also present. The dashboard offers four quick-action buttons, a column chart of reports by category, a pie chart of reports by type and a panel that opens the item reports.

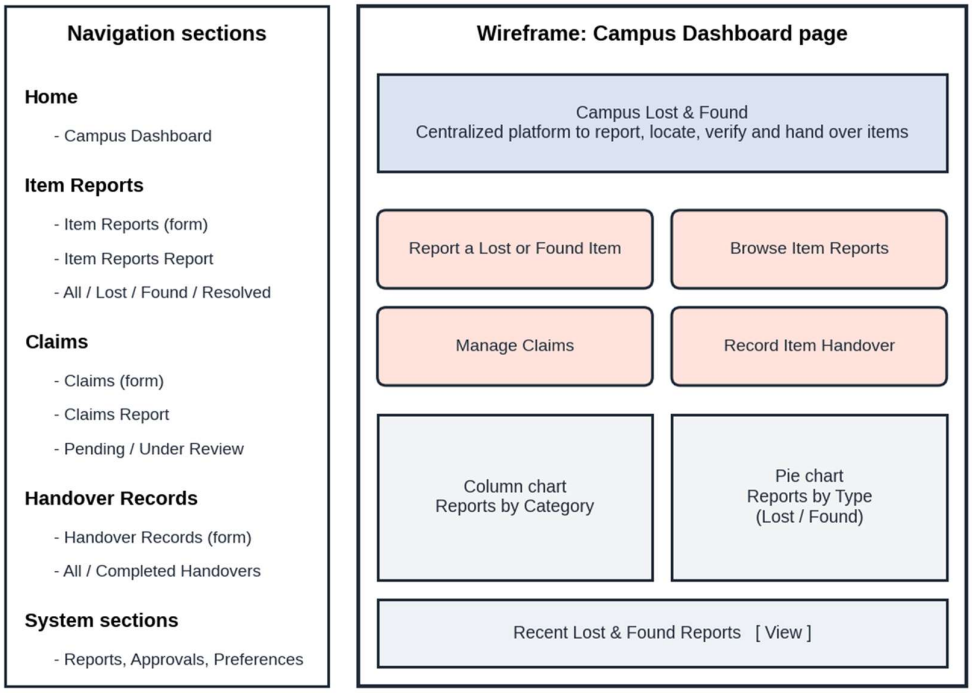


Figure 4.4: Navigation Structure and Wireframe of the Campus Dashboard

4.6 Access Control Design

Table 4.5: Role–Permission Matrix

Component	Administrator	Verifier (Admin Access)	Reporter / Claimant	Read-only user
Form: Item_Reports	A E V D	A E V	A V	V
Form: Claims	A E V D	A E V	A	V
Form: Handover_Records	A E V D	A E V	—	V
Reports: All, Lost, Found, Resolved items	V	V	V	V
Reports: Claims, Pending, Under Review	V	V	—	V
Reports: All and Completed Handovers	V	V	—	V
Page: Campus Dashboard	V	V	V	V

The matrix follows the principle of least privilege: reporters can add item reports and claims but cannot see handover records, and only verifiers and administrators can change claim status. Section 5.8 describes how the profiles were configured.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application LOST&FOUND was created in the Zoho Creator workspace with the date format dd-MMM-yyyy, the 24-hour time format and the UTC time zone. Its menu is organised into sections: Home (the Campus Dashboard page), Item Reports, Claims and Handover Records, each containing its form and reports. The sections All Orders and Add Order belong to the starter structure that Zoho Creator generates for a new application and are kept as a sample, together with the system sections Reports, Approvals and App Preferences. The Lato font, theme colour option 11 and a left-aligned application icon were chosen for the look and feel. The application was created on 19 October 2026 on the creator.zoho.in data centre. The complete application definition was exported as LOST_FOUND.ds (version 1.0, generated on 01-Nov-2026).

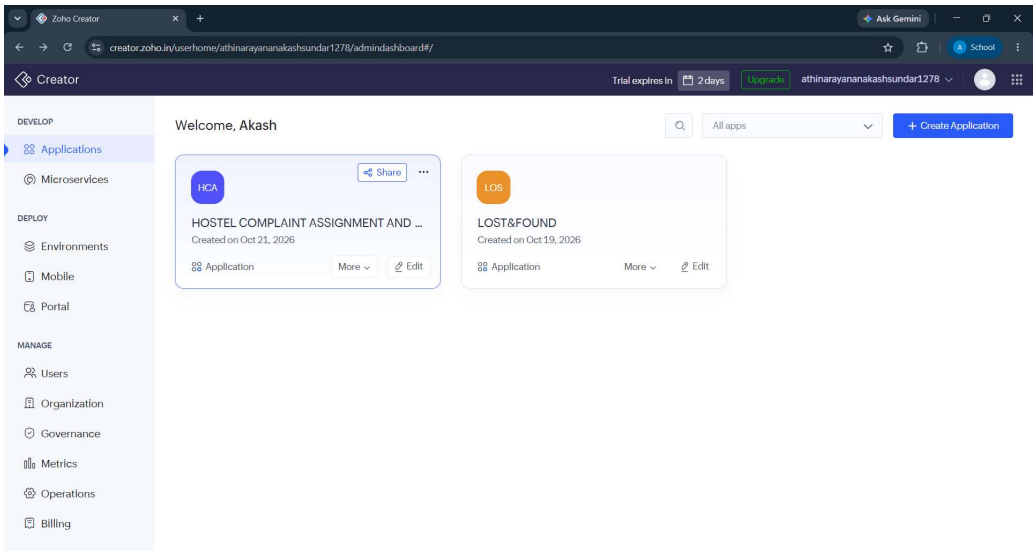


Figure 5.1: LOST&FOUND Application in the Zoho Creator Workspace

5.2 Forms and Data Capture

Item_Reports captures one lost or found item. It uses seven distinct field types: single line (item name, location, reporter name), radio buttons (report type), multi line (description), drop down (category and status), date, image (item photo) and e-mail (contact e-mail). Eight fields are mandatory. The photo field accepts a camera capture

or a gallery file, which suits mobile use. Status has the initial value Open, and the success message is “Data Added Successfully!”.

Claims captures an ownership claim. It uses a name field (marked as personal data), a lookup to Item_Reports that displays the item name, multi-line fields for ownership evidence and review notes, a drop-down claim status (Pending, Under Review, Approved, Rejected; initial value Pending) and a single-line field for the assigned verifier’s e-mail address.

Handover_Records captures the final handover. It has a lookup to Item_Reports, a lookup to Claims (Approved Claim), a mandatory Verified By field, a mandatory Handover Date, a Handover status drop down (Completed, Pending) and Resolution Notes. Across the three forms ten distinct field types are used in total when the Number field of the sample Add Order form is included.

All forms use left-aligned labels and the Submit button on add, and Update and Cancel buttons on edit. Conditional layout rules were not required because every field is relevant in every situation; field-level rules are implemented as mandatory flags and as Deluge validations (Section 5.5).

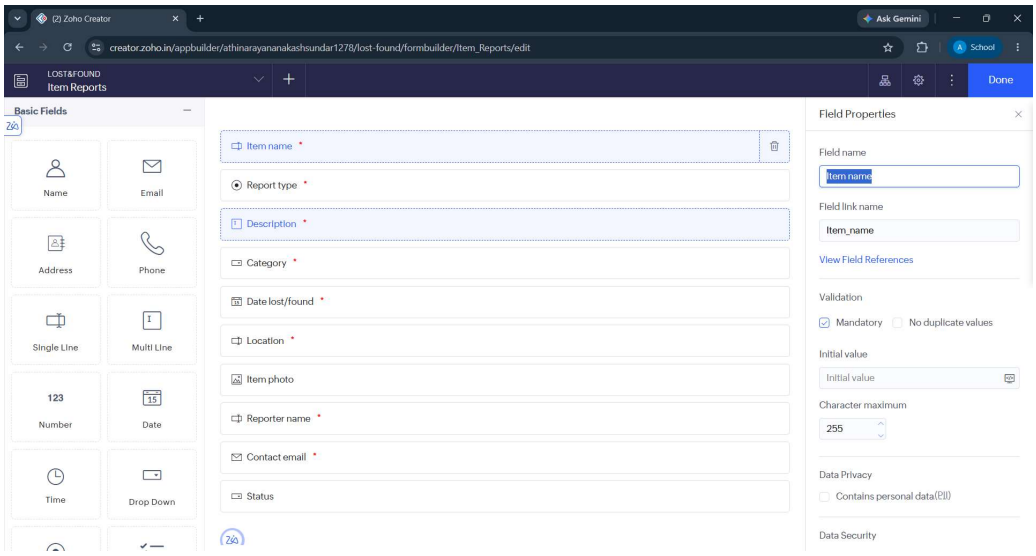


Figure 5.2: Item_Reports Form in the Form Builder

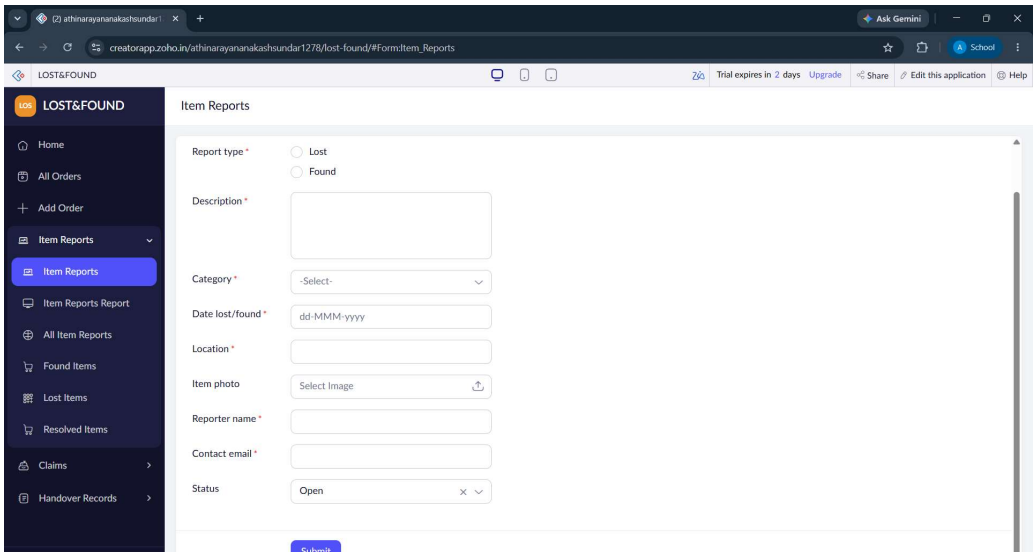


Figure 5.3: Item_Reports Form in Live Mode

5.3 Reports

All eleven reports are list reports (Table 5.1). Reports that show a subset of records carry a criteria on the source form, so the Lost, Found and Resolved items and the Pending, Under Review and Completed views are separate reports rather than one long list. Filters on Status and an "Exclude Resolved" filter help users narrow the data, and the item reports show the photograph as a thumbnail. The default reports provide a quick view with Edit, Duplicate and Delete actions. Conditional formatting is used only in the sample All Order Report. Kanban, calendar and pivot reports were not needed for this process; Section 8.5 discusses this choice.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Item Reports Report	List (default)	Item_Reports	All records; no filter	Quick view with Edit, Duplicate, Delete	Administrator, Verifier
All Item Reports	List	Item_Reports	Filter: Status; sort: Date lost/found descending; photo thumbnail	—	Administrator, Verifier

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Lost Items	List	Item_Reports	Criteria: Report type = Lost; filters: Status, Exclude Resolved; sort: date descending	—	Administrator, Verifier
Found Items	List	Item_Reports	Criteria: Report type = Found; filters: Status, Exclude Resolved; sort: date descending	—	Administrator, Verifier
Resolved Items	List	Item_Reports	Criteria: Status = Resolved; sort: date descending	—	Administrator
Claims Report	List (default)	Claims	All records; no filter	—	Administrator, Verifier
Pending Claims	List	Claims	Criteria: Claim Status = Pending; filter: Claim Status; sort: Added Time descending	—	Administrator, Verifier
Claims Under Review	List	Claims	Criteria: Claim Status = Under Review; sort: Added Time descending	—	Administrator, Verifier
All Handover Records	List (default)	Handover_Records	All records; no filter	—	Administrator, Verifier
Completed Handovers	List	Handover_Records	Criteria: Handover status = Completed; sort: Handover Date descending	—	Administrator, Verifier

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Order Report	List (sample)	Add_Order	Conditional formatting of Status; fixed column widths	—	Administrator

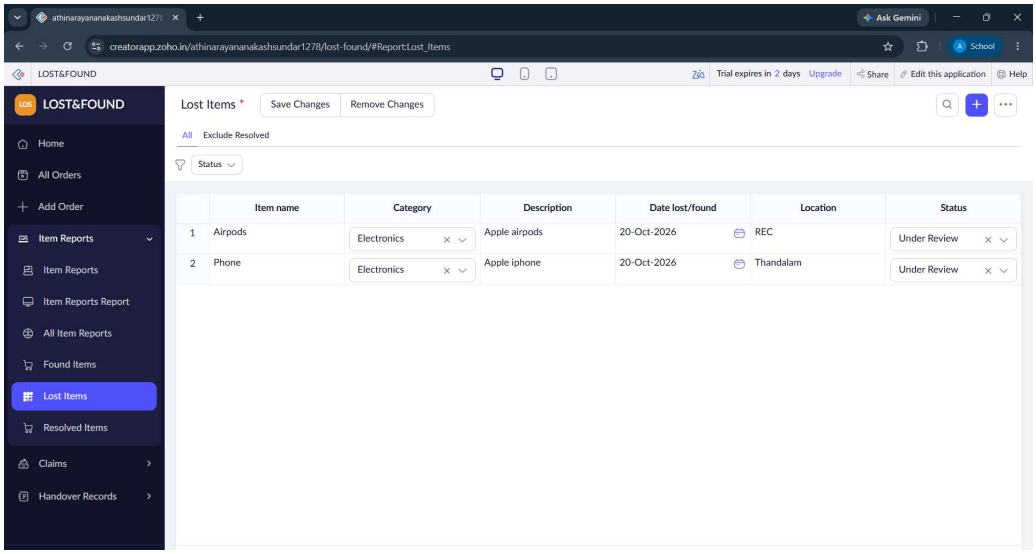


Figure 5.4: Lost Items Report

5.4 Pages and Dashboards

The Campus Dashboard page is built with the UI Builder and is meant for every role. It contains a heading ("Campus Lost & Found") with a one-line description of the platform; four buttons that open the Item Reports form, the Item Reports report, the Claims form and the Handover Records form; a column chart that counts item reports by Category (with data labels); a pie chart that counts reports by Report type with the legend at the top; and a panel titled "Recent Lost & Found Reports" with a View button that opens the item reports. The buttons open in a new window.

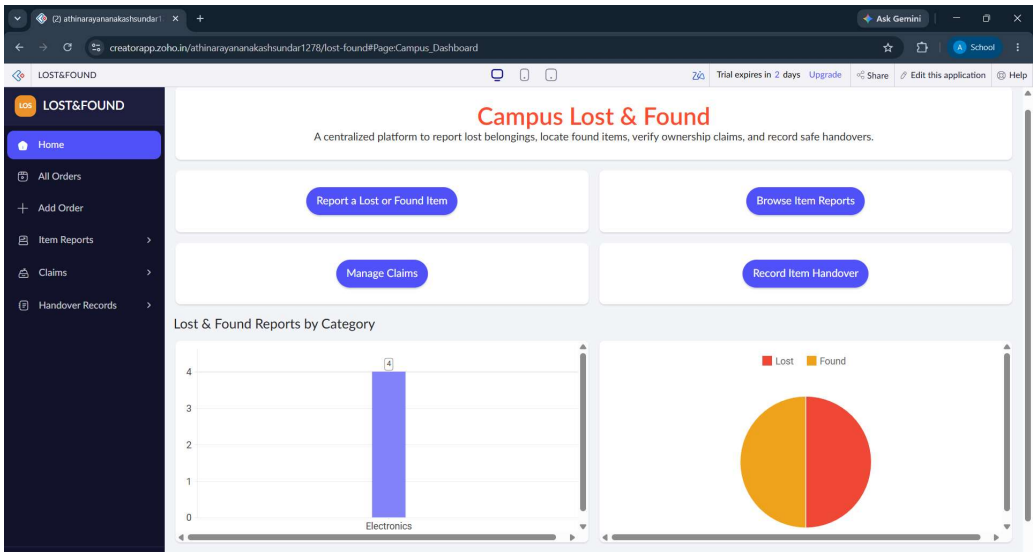


Figure 5.5: Campus Dashboard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Claims → On Validate	Block a claim if the item report is already Resolved	fetch with criteria, if, alert, cancel submit
DS-02	Claims → On Validate	Require an item report on every claim	isNull(), alert, cancel submit
DS-03	Claims → On Success	E-mail the assigned verifier about a new claim	if-else, sendmail, string concatenation, info
DS-04	Claims → On Load	Default Claim Status to Pending	isNull(), field assignment
DS-05	Claims → On Validate	Make ownership evidence compulsory	isEmpty(), field-level alert, cancel submit
DS-06	Handover_Records → On Validate	Allow a handover only for an existing, Approved claim	nested if / else if, fetch, count()
DS-07	Handover_Records → On Load	Default Handover status to Completed	field assignment
DS-08	Handover_Records → On Validate	Require an approved claim	isNull(), field-level alert
DS-09	Handover_Records → On Validate	Require an item report	isNull(), field-level alert
DS-10	Item_Reports → On Validate	Check the structure of the contact e-mail	contains(), toList(), size(), get(), isBlank(), for each, break

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-11	Item_Reports → On Success	E-mail the administrator and the reporter about a new report	Send Email action, merge fields

Code 5.1: Validation script DS-01 – block claims on resolved item reports

```
// DS-01 | Form: Claims | Event: On Validate
// Fetch the referenced Item Report record using the lookup field
itemReportRecord = Item_Reports[ID == input.Item_Report];
// Check if the record exists and its status is 'Resolved'
if(itemReportRecord.Status == "Resolved")
{
    alert "Cannot submit claim: The referenced Item Report is already
Resolved.";
    cancel submit;
}
```

Code 5.2: Notification script DS-03 – e-mail the assigned verifier

```
// DS-03 | Form: Claims | Event: On Success
// Fetch the assigned verifier's email from the input
assignedVerifierEmail = input.Assigned_Verifier;
// Check if the assigned verifier email is provided
if(!assignedVerifierEmail.isNull() && assignedVerifierEmail != "")
{
    // Send notification email to the assigned verifier
    sendmail
    [
        from :zoho.adminuserid
        to :assignedVerifierEmail
        subject :"New Claim Assigned for Review"
        message :"<p>Dear Verifier,</p><p>A new claim has been
submitted and assigned to you for review.</p><p>Claim
Details:</p><ul><li>Claimant Name: "
            + input.Claimant_name.first_name
            + " "
            + input.Claimant_name.last_name
            + "</li><li>Item: "
            + input.Item_Report.Item_name
            + "</li><li>Status: Pending</li></ul><p>Please log in to
review the claim at your earliest convenience.</p>"
        content type :HTML
    ]
}
else
{
    // Log if no verifier is assigned
    info "No verifier assigned for this claim. Notification skipped.";
}
```

DS-01 fetches the referenced item report by its ID and stops the submission with an alert if its status is Resolved, which prevents duplicate or invalid claims. DS-02 and DS-05 make sure that an item report is selected and that ownership evidence is present. DS-04 sets the claim status to Pending when the form loads.

DS-03 runs after a claim is saved. If an assigned verifier address is present it builds an HTML message from the claimant name and item name and sends it with sendmail; otherwise it writes a line to the log with info and skips the e-mail. DS-06 fetches the claim named in a handover and accepts the handover only if the claim exists and its status is Approved; DS-08 and DS-09 require the two lookups, and DS-07 sets the handover status to Completed.

DS-10 splits the contact e-mail at the @ sign, checks that both parts are present and that every part of the domain is non-empty, and shows an alert if the structure is invalid. The Email field type of the platform also validates the format, so DS-10 acts as an additional, explicit check. Complete listings are in Appendix C.

5.6 Workflows and Automation

All eleven workflows are form workflows. Schedule, approval, Blueprint and batch workflows were not implemented: claim review is a manual decision by the verifier and is tracked through the Claim Status field, and no time-based reminder was required. These options are discussed in Sections 8.4, 8.5 and 9.2.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Validate Item Report Status for Claim	Form Workflow	Claims, add or edit → On Validate	Item report status = Resolved	Alert; cancel submit	DS-01
Validate Claim Item Reference	Form Workflow	Claims, add or edit → On Validate	Item Report is empty	Alert; cancel submit	DS-02
Notify Verifier on New Claim	Form Workflow	Claims, add → On Success	Assigned Verifier is not empty	Send e-mail to verifier; otherwise log a message	DS-03
Set Default Claim Status	Form Workflow	Claims, add → On Load	Claim Status is empty	Set status to Pending	DS-04
Validate Claim Ownership Evidence	Form Workflow	Claims, add or edit → On Validate	Ownership Evidence is empty	Field alert; cancel submit	DS-05
Validate Claim Status for Handover	Form Workflow	Handover_Records, add or edit → On Validate	No claim, claim not found or claim not Approved	Alert; cancel submit	DS-06

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Set Default Handover Status	Form Workflow	Handover_Records, add → On Load	Always	Set status to Completed	DS-07
Validate Approved Claim Reference	Form Workflow	Handover_Records, add or edit → On Validate	Approved Claim is empty	Field alert; cancel submit	DS-08
Validate Item Report Reference	Form Workflow	Handover_Records, add or edit → On Validate	Item Report is empty	Field alert; cancel submit	DS-09
Validate Item Report Email	Form Workflow	Item_Reports, add or edit → On Validate	Contact e-mail is not empty	Check structure; alert if invalid	DS-10
Notify Admin – New Item Report	Form Workflow	Item_Reports, add → On Success	Always	Send e-mail to administrator and reporter	DS-11

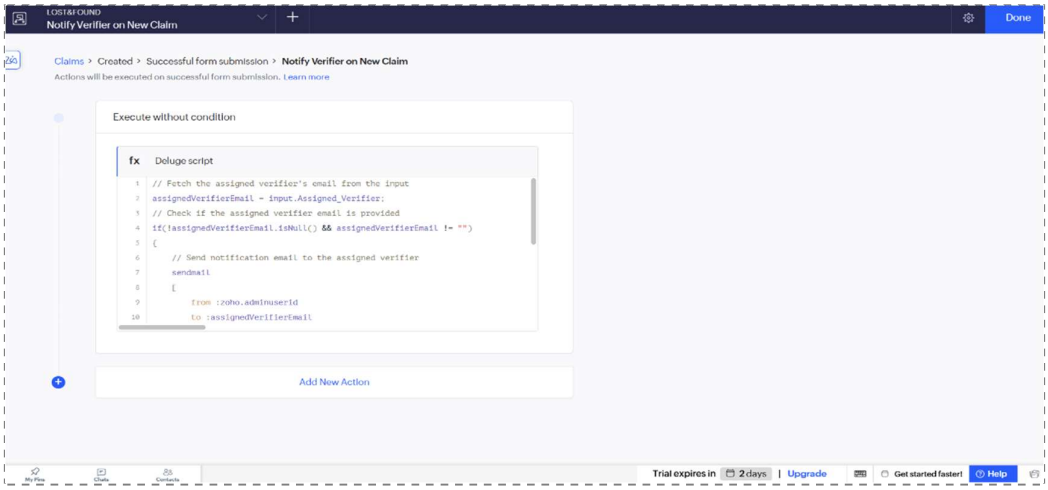


Figure 5.6: Notify Verifier on New Claim Workflow Configuration

5.7 Integrations

The only integration in this version is outbound e-mail through the mail service built into Zoho Creator. The received notification (Figure 5.7) comes from the Zoho Creator mail sender and lists the item name, report type, category and location. Both notifications use the application's administrator as the sender, so no password, key or token is stored in any script. No connection, invokeurl REST call, webhook or Creator REST API call was needed because all data stays inside the application; Section 9.2 proposes SMS and campus-system integrations as next steps.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Claim e-mail	Zoho Creator mail service; sender zoho.adminuserid	Deluge sendmail task (DS-03)	Platform-managed; no credentials in the script	Claim details sent to the assigned verifier
Item report e-mail	Zoho Creator mail service; sender zoho.adminuserid	Send Email action of the workflow builder (DS-11)	Platform-managed	Item details sent to the administrator and the reporter
External API / Zoho CRM	—	Not implemented in this version	—	Listed as a future enhancement

Code 5.3: Send Email action DS-11 – notification to the administrator and the reporter

```
// Script ID : DS-11
// Workflow  : Notify_Admin_New_Item_Rep
// Location  : Form Item_Reports → On Add → On Success
// Purpose   : E-mail the administrator and the reporter about a new
report
// Note      : configured with the workflow builder's Send Email
action
//             (no custom Deluge script); administrator address masked
from    : zoho.adminuserid
to      : <administrator e-mail address>, Contact_email
subject : "New Lost & Found Item Report Submitted"
message : A new item report has been submitted.
          Item name: ${Item_name}
          Report type: ${Report_type}
          Category: ${Category}
          Location: ${Location}
          Please open the LOST&FOUND application to review the report.
```

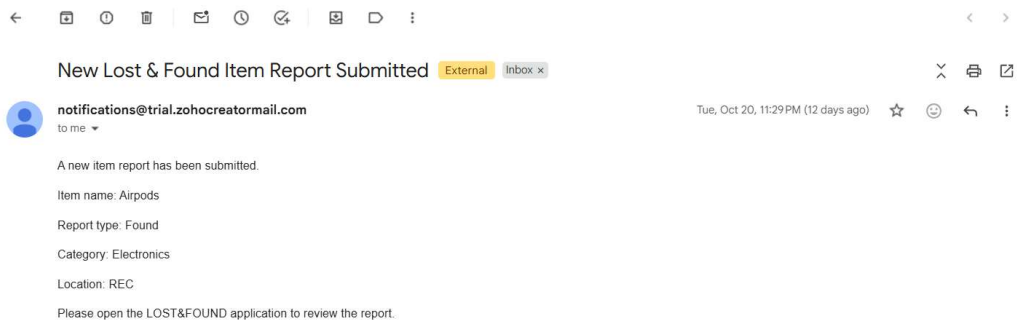


Figure 5.7: Notification E-mail Received After an Item Report Is Submitted

5.8 User Management and Role-Based Access

The access model uses the profiles and roles of Zoho Creator. The exported application defines five permission profiles: Administrator (all permissions), Read and Write (read or write access to all components), Admin Access (a custom profile for verifiers) and Customer (the default portal profile with add and view permission). The Admin Access profile enables creation and menu access for the Claims, Handover_Records and Item_Reports forms, makes all their fields visible and grants View and Edit on their reports (Pending Claims, Claims Report, Claims Under Review, All Handover Records, Completed Handovers, Item Reports Report, All Item Reports, Lost Items and Found Items). It also gives tab access to the Campus Dashboard. Resolved Items and the sample order report are therefore visible only to administrators.

A role hierarchy is defined with the role CEO at the top, whose users can access the data of all other users, and the child role Admin. Fields that hold personal data, the claimant name and the contact e-mail, are flagged as personal data, and the profiles carry the PII access permission. Record-level criteria and a configured customer portal are not part of this version; they would allow reporters to see only their own records and are listed in Section 9.2.

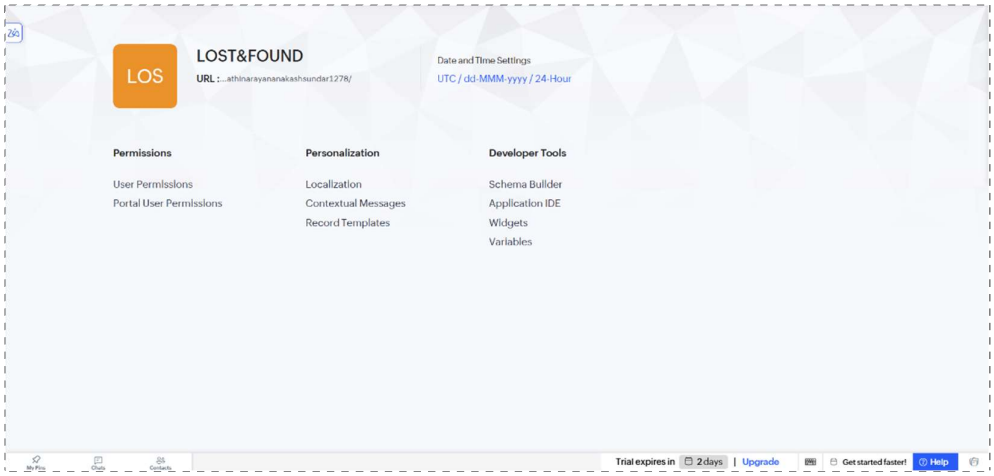


Figure 5.8: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Analytics is provided by two charts on the Campus Dashboard. The column chart counts item reports grouped by Category and the pie chart counts them by Report type (Lost or Found), both computed live from the Item_Reports form. They show administrators which categories of items are lost most often and the balance between

lost and found reports. In the sample data the column chart shows four item reports in the category Electronics and the pie chart shows an equal share of Lost and Found reports (Figure 5.9). The exported application contains no Zia or AI fields such as OCR, prediction or sentiment analysis.

[State whether any AI-assisted builder or script generation was used while developing the application, which parts it produced and how you verified them.]



Figure 5.9: Dashboard Charts (Analytics)

5.10 Pro-Code Extension

No pro-code extension was used. Every requirement of the application, namely validation, defaulting and notification, was met with form workflows and Deluge scripts, so Zoho Catalyst functions, JavaScript widgets or Creator REST API calls would have added complexity without benefit. Pro-code could be used later, for example in a Catalyst function that compares found-item photographs with lost-item descriptions, or in a widget that shows item locations on a campus map.

5.11 Mobile and Portal Access

The application can be opened in the Zoho Creator mobile app after sign-in. The Item photo field is configured with the primary camera, camera switching and gallery selection, so a person who finds an item can photograph it and file the report from a phone. The default Customer (portal) profile is present in the application definition; portal-specific configuration is not part of this version.

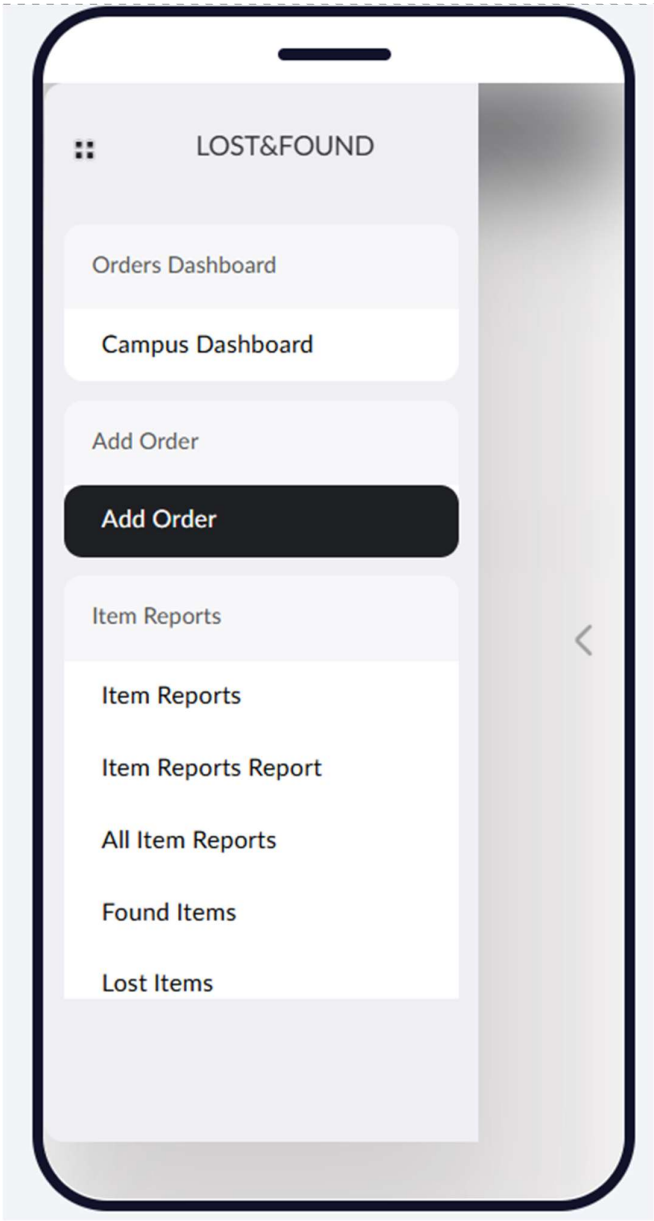


Figure 5.10: Application on the Zoho Creator Mobile App

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is organised at four levels. Unit testing checks each form and script on its own: mandatory fields, default values and every validation message. Integration testing follows a record through the forms, from an item report to a claim and finally a handover, and checks the e-mail notifications. Role-based testing signs in with an Administrator user and an Admin Access user and checks that each can open only the components allowed by the profile. User acceptance testing records the feedback of end users (Section 6.5).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit an item report with all mandatory fields	Open Item Reports; type Lost; fill all fields; Submit	Record created; message “Data Added Successfully!”	[ ]	[ ]
TC-02	FR-01	Submit an item report with Location empty	Leave Location blank; Submit	Mandatory-field error; record not saved	[ ]	[ ]
TC-03	FR-02	Enter a malformed contact e-mail	Enter “abc@domain”; Submit	Alert “Please enter a valid email address”	[ ]	[ ]
TC-04	FR-03	Check default status of a new item report	Open the Item Reports form	Status shows Open	Status shows Open on the new form (Figure 5.3)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-05	FR-04	E-mail on new item report	Submit a valid item report	E-mail “New Lost & Found Item Report Submitted” received by administrator and reporter	E-mail “New Lost & Found Item Report Submitted” received from the Zoho Creator mail sender with item name, type, category and location (Figure 5.7)	Pass
TC-06	FR-05	Lost Items report	Add Lost and Found items; open Lost Items	Only Lost items that are not Resolved, newest first	Lost Items report lists the lost items (Airpods, Phone), newest first (Figure 5.4)	Pass
TC-07	FR-05	Resolved Items report	Set an item to Resolved; open Found Items and Resolved Items	Item leaves Found/Lost Items and appears in Resolved Items	[ ]	[ ]
TC-08	FR-06	Submit a valid claim	Open Claims; choose item report; enter name and evidence; Submit	Claim saved with success message	[ ]	[ ]
TC-09	FR-07	Submit a claim without evidence	Leave Ownership Evidence blank; Submit	Alert “Ownership Evidence is required”; not saved	[ ]	[ ]
TC-10	FR-07	Claim on a resolved item report	Choose an item report with status Resolved; Submit	Alert “Cannot submit claim: The referenced Item Report is already Resolved.”	[ ]	[ ]

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-11	FR-08	Default claim status	Open the Claims form	Claim Status shows Pending	[ ]	[ ]
TC-12	FR-09	Verifier notification	Submit a claim with a verifier e-mail; repeat with the field empty	E-mail “New Claim Assigned for Review” sent; no e-mail and a log message when empty	[ ]	[ ]
TC-13	FR-10, FR-11	Move a claim to Under Review	Edit a pending claim; set Under Review	Claim leaves Pending Claims and appears in Claims Under Review	[ ]	[ ]
TC-14	FR-12	Handover for a claim that is not Approved	Select a Pending claim in Handover Records; Submit	Alert “The referenced claim is not approved. Handover cannot be submitted.”	[ ]	[ ]
TC-15	FR-12	Handover without an approved claim	Leave Approved Claim blank; Submit	Alert asking to select an approved claim	[ ]	[ ]
TC-16	FR-13	Valid handover	Approve a claim; add handover with verifier, date and notes	Record saved with status Completed; listed in Completed Handovers	[ ]	[ ]
TC-17	FR-14	Campus Dashboard	Open Campus Dashboard; click each button	Charts shown; buttons open the correct forms and report	Dashboard shows the four buttons, column chart and pie chart (Figures 5.5, 5.9)	Pass
TC-18	FR-15	Access by profile	Sign in as a user without the Admin Access profile; open Handover Records	Component not available to the user	[ ]	[ ]

Test descriptions and expected results are derived from the scripts and forms of the application. Results are filled in where screenshot evidence exists; the remaining Actual Result and Status cells are completed after each test is executed.

6.3 Debugging Deluge Scripts

Deluge scripts are debugged with three techniques. First, alert statements show a message to the user and cancel the submission, which makes validation failures visible. Second, DS-03 uses an info statement to write a line to the log when no verifier is assigned, so a skipped notification can be explained. Third, the script editor's execute and error messages and the workflow logs of Zoho Creator show which line failed. Testing each script separately before enabling the next one kept the cause of a failure easy to find.

[Describe one real bug you met during development, how you found it and how you fixed it.]

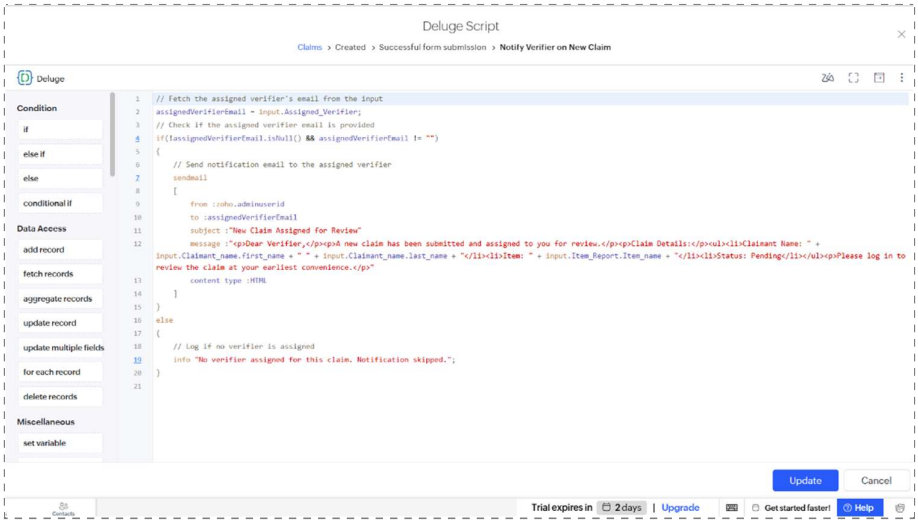


Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
[BUG-01]	[ ]	[ ]	[ ]	[Closed]
[ ]	[ ]	[ ]	[ ]	[ ]

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides separate Development, Stage and Production environments so that changes can be built and tested before they reach live users. The deployment strategy for LOST&FOUND is to build and test the forms, scripts and reports in Development, run the test cases of Table 6.1, remove sample data, configure users and profiles in Production and then publish the application. The application definition exported as LOST_FOUND.ds (version 1.0, 01-Nov-2026) is kept as a snapshot that can be used to restore the application if a release has to be rolled back.

[State which environments you actually used, the publish date and the version notes recorded.]

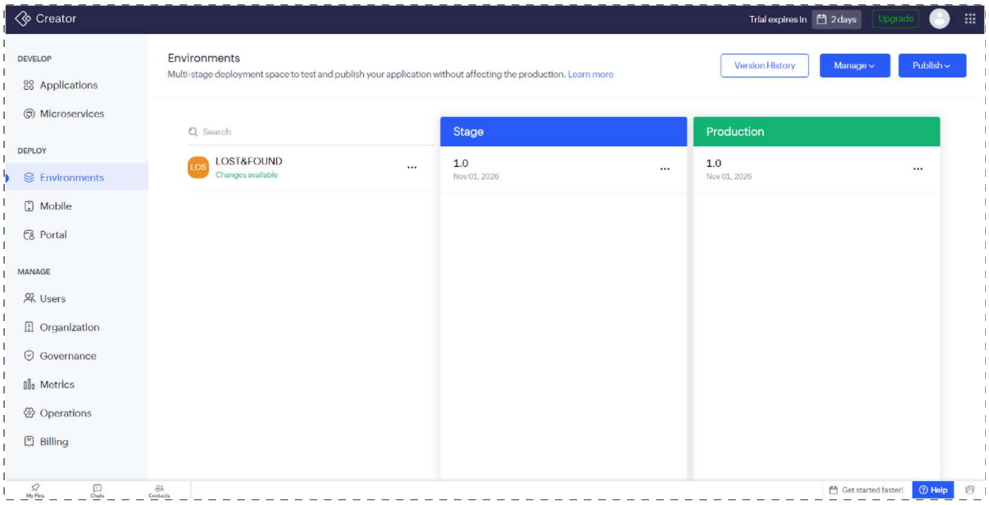


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	[ ]
2	Sample/test data removed or isolated	[ ]
3	Roles, permissions and users configured in Production	[ ]
4	Connections authorised in Production	[ ]
5	Schedules and workflows enabled	[ ]
6	Application published and URL verified	[ ]
7	Version / release notes recorded	[ ]

7.2 Performance Optimisation

Performance was designed in rather than tuned afterwards. Reports that show a subset of records carry criteria on the source form, so each loads only the records it needs. The Lost, Found and Resolved reports exclude or include resolved items by default, item photographs are shown as thumbnails in lists, and report columns are limited to the fields needed. In scripts, DS-01 and DS-06 fetch a single record by its ID instead of scanning a form, no fetch is placed inside a loop (the only loop in the application, in DS-10, iterates over the parts of an e-mail domain), and invalid submissions are cancelled before they are saved. Timings were not recorded, so Table 7.2 describes the effect qualitatively.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Criteria on report source	Lost, Found, Resolved, Pending, Under Review and Completed reports	All records loaded in one list	Only matching records loaded	Fewer records per page load
Single-record fetch by ID	DS-01, DS-06	Scan of the whole form	One record read per validation	Less data read per submission
Thumbnails for photographs	Item report lists	Full-size images in lists	Thumbnail dimension	Lighter pages
Default Exclude Resolved filter	Lost Items, Found Items	Closed cases remain in the list	Closed cases hidden by default	Shorter working lists
Cancel invalid submissions early	DS-01 to DS-09	Invalid records saved and corrected later	Rejected before saving	No clean-up of bad data

7.3 Security Features

Security is based on permissions, personal-data flags and validation rules. An encrypted field was not used because the application stores no highly sensitive identifiers such as card or government ID numbers, and no OAuth connection is needed because there is no external integration.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Profile-based permissions	Administrator, Admin Access, Read, Write and Customer profiles	Unauthorised access to forms, reports and pages
Module permissions for verifiers	Admin Access: Claims, Handover_Records, Item_Reports, their reports and Campus_Dashboard	Over-privileged access by verifiers
Personal-data flag	Claimant_name (Claims), Contact_email (Item_Reports)	Exposure of personal information
Compulsory ownership evidence and verifier review	Claims form, DS-05, Claim_Status	Fraudulent claims on found items
Server-side validation scripts	DS-01, DS-06, DS-08, DS-09	Invalid state changes such as a claim on a resolved item or a handover without approval
Platform sender, no stored credentials	DS-03 and DS-11 use zoho.adminuserid	Credential leakage
Role hierarchy	CEO → Admin	Uncontrolled visibility of other users' data

7.4 Governance and Administration

Zoho Creator records the user and time of every record change, and the Added Time system field is used to order the claim reports; the platform audit trail and administration settings let the administrator review changes. Components follow a naming convention: underscore-separated link names for forms (Item_Reports, Claims, Handover_Records) and descriptive workflow names beginning with Validate, Set Default or Notify. The exported .ds file serves as a backup and version snapshot. The administrator manages user licences, profiles and roles and is responsible for maintaining categories and statuses, while verifiers maintain claim and handover records.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario follows one found item through all roles. A student finds a bunch of keys in the library and files a Found item report with a photograph (Figure 8.1); the administrator and the reporter receive an e-mail. The owner sees the item in the Found Items report and submits a claim with ownership evidence and the verifier's e-mail; the verifier is notified, reviews the claim and approves it (Figure 8.2). The verifier then records the handover and closes the item report as Resolved, and the dashboard charts and the Completed Handovers report reflect the result (Figure 8.3).

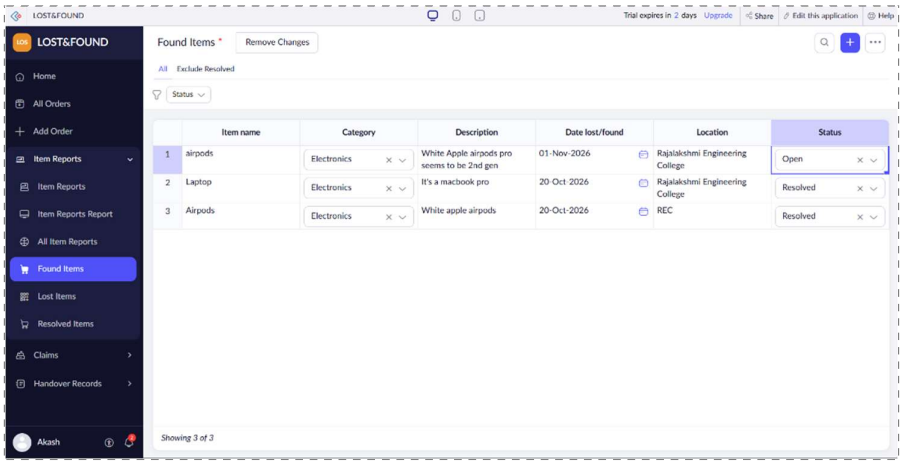


Figure 8.1: Scenario Step 1 – Found Item Reported

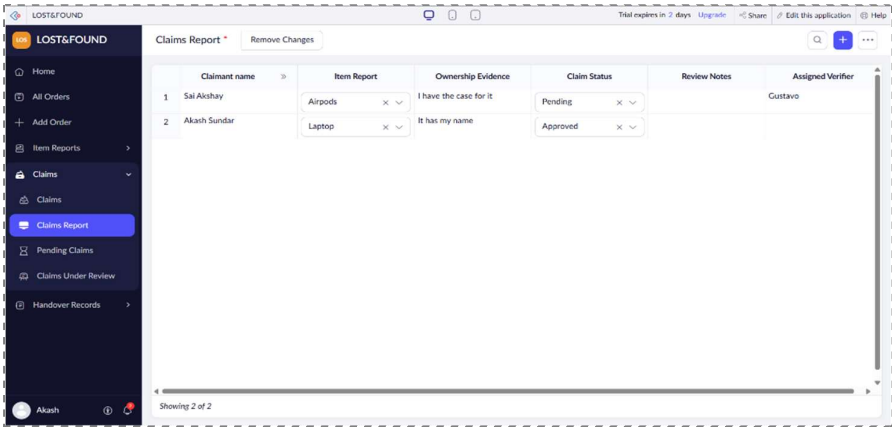


Figure 8.2: Scenario Step 2 – Claim Submitted and Verified

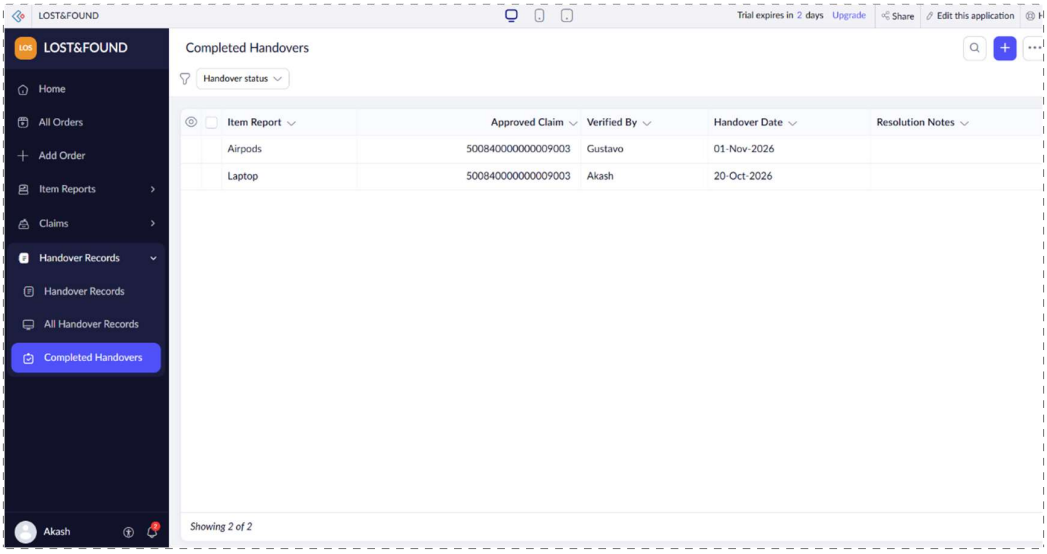


Figure 8.3: Scenario Step 3 – Handover Recorded

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Design a normalised data model and forms	Sections 4.2, 4.3, 5.2	Three linked forms (Item_Reports, Claims, Handover_Records) with lookups; Figures 4.2, 5.2, 5.3
Automate validation, defaults and notification	Sections 5.5, 5.6	Eleven form workflows and ten Deluge scripts; Tables 5.2, 5.3; Codes 5.1, 5.2
Role-based dashboards and reports	Sections 5.3, 5.4, 5.8	Eleven list reports, Campus Dashboard with two charts, Admin Access profile; Figures 5.4, 5.5, 5.8
E-mail notifications through the Zoho mail service	Section 5.7	DS-03 and DS-11; Code 5.2, Code 5.3, Figure 5.7
Secure deployment with governance	Sections 7.1 to 7.4	Environments, security controls and governance practices; Tables 7.1, 7.3; Figures 7.1, 7.2

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Manual look-up in a register or by enquiry	Filtered reports and one record per case	Faster retrieval and no manual look-up
Manual steps	Writing, searching, verbal checking and signing	Data entered once; status and e-mails are automatic	Fewer manual steps
Ownership verification	Verbal description	Written evidence, reviewed claim and approved status before handover	Recorded, auditable decision
Notifications	None	E-mail to administrator, reporter and verifier	Automatic communication
Report generation	Manual counting	Dashboard charts by category and type and filtered reports	Instant summaries
Data accessibility	Desk-bound registers and messages	Web and mobile access with role-based control	Available anywhere to authorised users

The comparison is qualitative; no stopwatch or counting study of the old and new process was carried out.

8.4 Limitations of the Low-Code Approach

Several constraints of the platform were met. The user interface is limited to the components Zoho Creator offers, so the dashboard and report layouts could not be freely designed. Logic is attached to form events, so an update across forms must be written explicitly; for example, completing a handover does not yet mark the item report as Resolved or close the claim, and the verifier does this by hand. Validation feedback is limited to plain alerts. Record, API-call and storage limits depend on the subscription edition, and the application can run only on Zoho Creator, which is a form of vendor lock-in even though the definition can be exported. These limits were handled by keeping rules simple, validating early with Deluge scripts and documenting the manual steps in the user manual.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the no-code → low-code → pro-code spectrum, most of the application is no-code configuration: the three forms and their field properties, eleven reports, the dashboard page with its buttons and charts, and the profiles and roles. Low-code is needed for the Deluge scripts: validations, default values and the verifier e-mail, plus the Send Email action configured in the workflow builder. Pro-code would be needed only for features beyond the platform's components, such as image matching of lost and found items or integration with the campus student system, and I would move to pro-code only when a requirement could not be met by a form event or a built-in integration.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project delivered Lost & Found Management, a Zoho Creator application that moves the campus lost-and-found process from registers and messages to a structured, verified workflow. Three linked forms capture item reports, ownership claims and handovers; eleven list reports and a Campus Dashboard make the data visible; eleven form workflows validate input, set defaults and send e-mail notifications; and profiles, roles and personal-data flags control access.

The work shows that a small, workflow-centred institutional service can be built with configuration and a few short Deluge scripts, without a hosted back end or a separate mobile app. The main gaps are manual steps between forms and the absence of external integrations, which are addressed in the future enhancements.

9.2 Future Enhancements

- Mark the item report as Resolved and close the claim automatically when a handover is completed, using an on-success workflow on Handover_Records.
- Add an approval process or Blueprint for claims and a scheduled workflow that reminds verifiers of claims pending for several days.
- Use Zia or AI image features to suggest matches between found photographs and lost descriptions, and OCR to read identity cards.
- Send SMS or WhatsApp notifications and integrate with the campus identity or student database through REST API or Zoho Catalyst.
- Configure a customer portal with record-level access so that reporters can track their own reports and claims, and add a Zoho Analytics view for trend analysis.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling with lookup fields, form and report design, Deluge scripting, workflow automation, role-based access and deployment on a low-code platform. These skills relate directly to roles such as citizen developer, low-code developer, solutions consultant and automation engineer, where business processes are turned into working applications quickly.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	LOST&FOUND menu: Home, Item Reports, Claims, Handover Records	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Item_Reports, Claims, Handover_Records (9 types; 10 with Add_Order)	5.2	Tables 4.2–4.4; Fig. 5.2
II	Lookup field (relationship between forms)	Y	Claims.Item_Report; Handover_Records.Item_Report, Approved_Claim	4.2, 5.2	Figure 4.2
II	Subform (parent–child data)	Y	Implemented	4.2, 5.2	—
II	Field validations / layout rules	Y	Mandatory fields; DS-01, 02, 05, 06, 08, 09, 10 (no layout rules)	5.2	Code 5.1
II	List report with filter, grouping, custom action	Partial	Filters and sorting in item and claim reports; no grouping or custom action	5.3	Figure 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Implemented	5.3	—
II	Pivot table / chart report	N	Charts are on the dashboard page, not chart reports	5.3	—
II	Page / dashboard built with UI Builder	Y	Page Campus_Dashboard	5.4	Figure 5.5
II	Deluge – form events (validate / success / field actions)	Y	On Validate, On Load, On Success events	5.5	Codes 5.1, 5.2
II	Deluge – functions, fetch, loops, collections	Partial	fetch (DS-01, DS-06), for each and lists (DS-10); no user-defined function	5.5	Appendix C
II	Email / in-app notifications	Y	DS-03 sendmail; DS-11 Send Email action	5.5, 5.6	Codes 5.2, 5.3
II	Form workflow	Y	Eleven form workflows	5.6	Table 5.3
II	Scheduled workflow	Y	Implemented	5.6	—
II	Approval process	Y	Implemented	5.6	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Blueprint	N	Not used	5.6	—
II	Batch / report workflow	Y	Implemented	5.6	—
II	Testing & debugging evidence	[Y / N]	Table 6.1 test cases	6	Figure 6.1
III	Integration with a Zoho service	Y	Implemented	5.7	—
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	Admin Access profile; roles CEO and Admin	5.8	Figure 5.8
III	Record-level access criteria / customer portal	N	Not configured in this version	5.8	—
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Column and pie charts on Campus_Dashboard	5.9	Figure 5.9
IV	Performance optimisation applied & measured	Partial	Applied (criteria reports, thumbnails, ID fetch); not measured numerically	7.2	Table 7.2
IV	Deployment across environments	[Y / N]	Development to Production	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Profiles, permissions and personal-data flags; no encrypted field or OAuth	7.3	Table 7.3
IV	Governance (audit trail, administration)	[Y / N]	Audit trail and administration settings	7.4	Figure 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	[Y / N]	Zoho Creator mobile app	5.11	Figure 5.10

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	[ ]	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	[ ]	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	[ ]	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	[ ]	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	[ ]	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Claims form – DS-01 to DS-05

```
// Script ID : DS-01
// Workflow  : Validate_Item_Report_Status_For_Claim
// Location  : Form Claims → On Add/Edit → On Validate
// Purpose   : Block a claim if the referenced Item Report is already
Resolved
// Fetch the referenced Item Report record using the lookup field
itemReportRecord = Item_Reports[ID == input.Item_Report];
// Check if the record exists and its status is 'Resolved'
if(itemReportRecord.Status == "Resolved")
{
    alert "Cannot submit claim: The referenced Item Report is already
Resolved.";
    cancel submit;
}

// Script ID : DS-02
// Workflow  : Validate_Claim_Item_Reference
// Location  : Form Claims → On Add/Edit → On Validate
// Purpose   : Ensure an Item Report is selected before a claim is
submitted
if(input.Item_Report.isNull())
{
    alert "Item Report is required to submit this claim";
    cancel submit;
}

// Script ID : DS-03
// Workflow  : Notify_Verifier_On_New_Claim
// Location  : Form Claims → On Add → On Success
// Purpose   : E-mail the assigned verifier when a new claim is
submitted
// Fetch the assigned verifier's email from the input
assignedVerifierEmail = input.Assigned_Verifier;
// Check if the assigned verifier email is provided
if(!assignedVerifierEmail.isNull() && assignedVerifierEmail != "")
{
    // Send notification email to the assigned verifier
    sendmail
    [
        from :zoho.adminuserid
        to :assignedVerifierEmail
        subject :"New Claim Assigned for Review"
        message :"<p>Dear Verifier,</p><p>A new claim has been
submitted and assigned to you for review.</p><p>Claim
Details:</p><ul><li>Claimant Name: "
            + input.Claimant_name.first_name
            + " "
            + input.Claimant_name.last_name
            + "</li><li>Item: "
            + input.Item_Report.Item_name
            + "</li><li>Status: Pending</li></ul><p>Please log in to
review the claim at your earliest convenience.</p>"
        content type :HTML
    ]
}
else
{
```

```
    // Log if no verifier is assigned
    info "No verifier assigned for this claim. Notification skipped.";
}

// Script ID : DS-04
// Workflow  : Set_Default_Claim_Status
// Location  : Form Claims → On Add → On Load
// Purpose   : Default the Claim Status to Pending on a new claim
// Set Claim Status to Pending when a new claim is created
if(input.Claim_Status.isNull())
{
    input.Claim_Status = "Pending";
}

// Script ID : DS-05
// Workflow  : Validate_Claim_Ownership_Evidence
// Location  : Form Claims → On Add/Edit → On Validate
// Purpose   : Make Ownership Evidence compulsory
if(input.Ownership_Evidence.isEmpty())
{
    alert(Ownership_Evidence,"Ownership Evidence is required");
    cancel submit;
}
```

C.2 Handover_Records form – DS-06 to DS-09

```
// Script ID : DS-06
// Workflow  : Validate_Claim_Status_For_Handover
// Location  : Form Handover_Records → On Add/Edit → On Validate
// Purpose   : Allow a handover only against an existing, Approved
claim
if(input.Approved_Claim == null)
{
    alert "Please select an approved claim before submitting the
handover.";
    cancel submit;
}
else
{
    claimRecord = Claims[ID == input.Approved_Claim];
    if(claimRecord.count() == 0)
    {
        alert "No claim record was found. Please select a valid
claim.";
        cancel submit;
    }
    else if(claimRecord.Claim_Status != "Approved")
    {
        alert "The referenced claim is not approved. Handover cannot
be submitted.";
        cancel submit;
    }
}

// Script ID : DS-07
// Workflow  : Set_Default_Handover_Status
// Location  : Form Handover_Records → On Add → On Load
// Purpose   : Default the Handover status to Completed on a new
handover record
// Set the Handover_status field to "Completed" when the record is
successfully submitted
```

```
input.Handover_status = "Completed";

// Script ID : DS-08
// Workflow  : Validate_Approved_Claim_Reference
// Location  : Form Handover_Records → On Add/Edit → On Validate
// Purpose   : Ensure an approved claim is selected
if(input.Approved_Claim.isNull())
{
    alert(Approved_Claim,"Please select an approved claim before
submitting.");
    cancel submit;
}

// Script ID : DS-09
// Workflow  : Validate_Item_Report_Reference
// Location  : Form Handover_Records → On Add/Edit → On Validate
// Purpose   : Ensure an Item Report is selected
if(input.Item_Report.isNull())
{
    alert(Item_Report,"Item Report is required to proceed with
handover");
    cancel submit;
}
```

C.3 Item_Reports form – DS-10 and DS-11

```
// Script ID : DS-10
// Workflow  : Validate_Item_Report_Email
// Location  : Form Item_Reports → On Add/Edit → On Validate
// Purpose   : Check the structure of the contact e-mail address
if(!input.Contact_email.isNull())
{
    emailValue = input.Contact_email;
    isValidFormat = true;
    if(emailValue.contains("@"))
    {
        parts = emailValue.toList("@");
        if(parts.size() == 2)
        {
            localPart = parts.get(0);
            domainPart = parts.get(1);
            if(localPart.isBlank() || domainPart.isBlank())
            {
                isValidFormat = false;
            }
            else if(!domainPart.contains("."))
            {
                isValidFormat = false;
            }
            else
            {
                domainParts = domainPart.toList(".");
                if(domainParts.size() < 2)
                {
                    isValidFormat = false;
                }
                else
                {
                    for each  domainSegment in domainParts
                    {
                        if(domainSegment.isBlank())
```

```
                        {
                            isValidFormat = false;
                            break;
                        }
                    }
                }
            }
        }
        else
        {
            isValidFormat = false;
        }
    }
    else
    {
        isValidFormat = false;
    }
    if(!isValidFormat)
    {
        alert(Contact_email,"Please enter a valid email address");
    }
}

// Script ID : DS-11
// Workflow  : Notify_Admin_New_Item_Rep
// Location  : Form Item_Reports → On Add → On Success
// Purpose   : E-mail the administrator and the reporter about a new
report
// Note      : configured with the workflow builder's Send Email
action
//             (no custom Deluge script); administrator address masked
from    : zoho.adminuserid
to      : <administrator e-mail address>, Contact_email
subject : "New Lost & Found Item Report Submitted"
message : A new item report has been submitted.
          Item name: ${Item_name}
          Report type: ${Report_type}
          Category: ${Category}
          Location: ${Location}
          Please open the LOST&FOUND application to review the report.
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the published application URL (see Project at a Glance) in a web browser and sign in with the Zoho account created for you by the administrator. On a phone, install the Zoho Creator app, sign in with the same account and open LOST&FOUND. The Campus Dashboard opens first; every other section is reached from the menu.

D.2 Instructions for Reporter / Claimant

- On the Campus Dashboard click “Report a Lost or Found Item”, select Lost or Found, enter the item name, description, category, date, location, your name and contact e-mail, add a photograph if you have one, and click Submit.
- To look for your item, click “Browse Item Reports” or open Lost Items or Found Items from the Item Reports menu.
- If an item is yours, open Claims, choose the Item Report, enter your name and the proof of ownership (for example marks, contents or receipt), enter the verifier’s e-mail if you know it and click Submit.

D.3 Instructions for Verifier and Administrator

- Open Claims → Pending Claims to see new claims, open a claim and check the Ownership Evidence.
- Edit the claim: set Claim Status to Under Review, add Review Notes and finally choose Approved or Rejected.
- When an approved claimant collects the item, open Handover Records, select the Item Report and the Approved Claim, enter Verified By, Handover Date and Resolution Notes, and click Submit.
- Open the Item Reports report, edit the item report and set its Status to Resolved. Completed handovers can be reviewed in Completed Handovers.
- Administrators additionally manage users, profiles and roles in the application settings and monitor the Campus Dashboard.